Ruoxi Li
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Skye Hall, Riverside, CA

APPOINTMENTS

2026–Present Math Fellow, University of California, Riverside

EDUCATION

2019–2026 **PhD**, University of Pittsburgh (Advisor: Prof. Roman Fedorov)
2016-2018 **MS**, University of Illinois Urbana-Champaign
2012-2016 **BS**, Nankai University

RESEARCH INTERESTS

My research focuses on algebraic geometry and representation theory. More specifically, I am interested in moduli stacks of Higgs bundles and bundles with connections, character varieties, and moduli spaces of quiver representations. Recently, I have also become interested in questions surrounding symplectic resolutions.

PUBLICATIONS AND PREPRINTS

1. Explicit formulas for mixed Hodge polynomials of character varieties of nilpotent groups (with Rahul Singh). To appear in *Asian Journal of Mathematics*. arXiv: 2410.10008.
2. Motivic classes of stacks in finite characteristic and applications to stacks of Higgs bundles. arXiv: 2509.09861.
3. Invariant algebraic connections on connected reductive groups (with Rudrendra Kashyap). *Algebras and Representation Theory*. Journal. arXiv.

PROJECTS

2022– Motivic classes of stacks in finite characteristic and applications to stacks of bundles with connections (joint with Roman Fedorov).
2024– Motives of moduli spaces of stable quiver representations (joint with Sergey Mozgovoy, Markus Reineke and Rahul Singh).

SEMINAR TALKS

11/2025 M-Seminar, Kansas State University.
03/2025 Arithmetic Seminar, Binghamton University.
11/2024 Graduate Student Seminar, University of Pittsburgh.
10/2024 Graduate Student Seminar, University of Pittsburgh.
09/2024 Algebraic Groups and the Geometric Langlands Program Seminar, University of Pittsburgh.

CONFERENCE TALKS (CONTRIBUTED)

08/2025 Workshop on Character Varieties and Higgs Bundles, University of Costa Rica. 10-minute Talk and poster session.

05/2025 GTA Philadelphia, Temple University. 30-minute Talk.
11/2024 Western Algebraic Geometry Symposium, University of Arizona. Poster session.
11/2024 Algebraic Geometry Northeastern Series, Dartmouth College. Poster session.
10/2024 BUGCAT Conference, Binghamton University. 20-minute Talk.
08/2024 Richmond Geometry Meeting, Virginia Commonwealth University. Poster session.

SERVICE AND PROFESSIONAL ACTIVITIES

2025 **Mentor**, CMU-Pitt Directed Reading Program
2024– **Referee**, Pittsburgh Interdisciplinary Mathematics Review
2020 **Co-organizer**, Seminar on Moduli of Vector Bundles, online

HONORS AND AWARDS

2019 Arts & Sciences Graduate Fellowship, University of Pittsburgh
2012 Freshman's Second Prize Scholarship, Nankai University

TEACHING EXPERIENCES

University of California, Riverside

Instructor

MATH 011 Introduction to Discrete Structures, FA 2026
MATH 022 Calculus for Business, FA 2026

University of Pittsburgh

Instructor

MATH 0290 Differential Equations, SU 2024

Teaching Assistant

MATH 0120 Business Calculus, FA 2020, 2021, SU 2025
MATH 0220 Analytic Geometry & Calculus 1, SP 2021, 2022, 2024
MATH 0230 Analytic Geometry & Calculus 2, SP 2021, 2025, SU 2021, FA 2021, 2022
MATH 0413 Intro Theoretical Mathematics, SU 2021
MATH 0420 Intro Theory 1-Variable Calculus, SP 2023
MATH 0430 Intro Abstract Algebraic Systems, SP 2023, 2024

Grader

MATH 0290 Differential Equations, FA 2024, SP 2026
MATH 0400 Finite Mathematics, SP 2021
MATH 1020 Applied Elementary Number Theory, FA 2022, 2023
MATH 1050 Combinatorial Mathematics, FA 2022, 2023
MATH 1470 Partial Differential Equations 1, FA 2021

CONFERENCES ATTENDED

10/2026 Geometric Representation Theory and 3d Mirror Symmetry, Simons Laufer Mathematical Sciences Institute (SLMath).
07/2026 16th Southeastern Lie Theory Workshop - ICM 2026 Satellite Conference, University of Virginia.
08/2025 Workshop on Character Varieties and Higgs Bundles, University of Costa Rica.

05/2025	Hodge Theory and Algebraic Geometry, Simons Center for Geometry and Physics, Stony Brook University.
05/2025	GTA Philadelphia, Temple University.
11/2024	Western Algebraic Geometry Symposium, University of Arizona.
11/2024	Algebraic Geometry Northeastern Series, Dartmouth College.
10/2024	BUGCAT Conference, Binghamton University.
08/2024	Richmond Geometry Meeting, Virginia Commonwealth University.
07/2023	WARTHOG workshop on the topic of "Categorified Coulomb branches", University of Oregon.
04/2023	Categorical methods in moduli theory, University of Pennsylvania.
09/2022	Workshop on "Moduli, motives, and bundles - New trends in algebraic geometry", online.
07/2022	Derived Categories, Moduli Spaces, and Hyperkähler Varieties, University of Michigan.
07/2021	GAeL (Géométrie Algébrique en Liberté) XXVIII, online.
05/2021	Special Month on Singularities and K-Stability, online.
06/2020	Overview of Springer theory, online.
06/2020	Workshop on Lie Theory and Integrable Systems in Symplectic and Poisson Geometry, online.
06/2020	Warwick virtual M2 Workshop, online.
05/2020	Geometry and Dynamics of Foliations, online.

Last updated: October 23, 2026